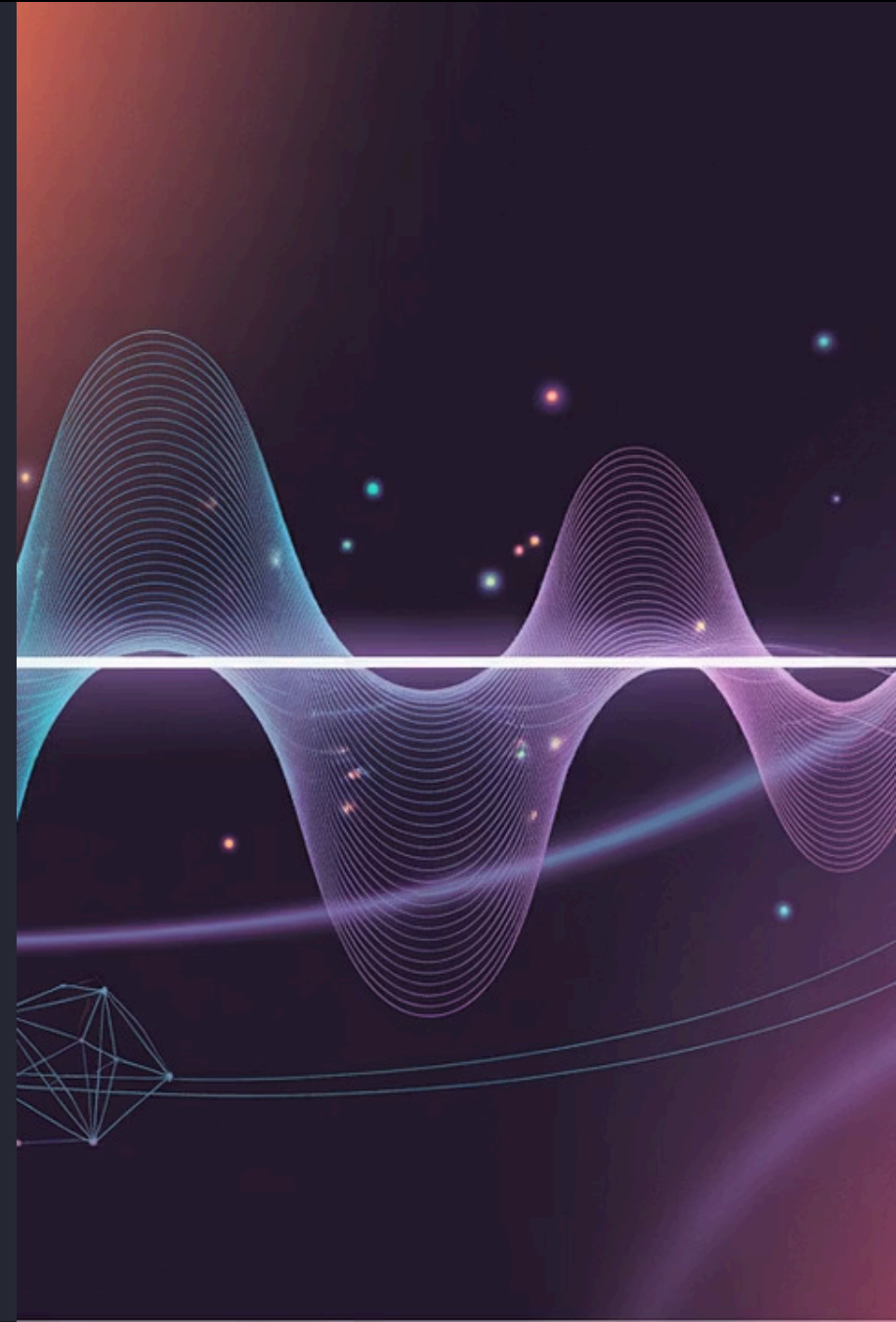
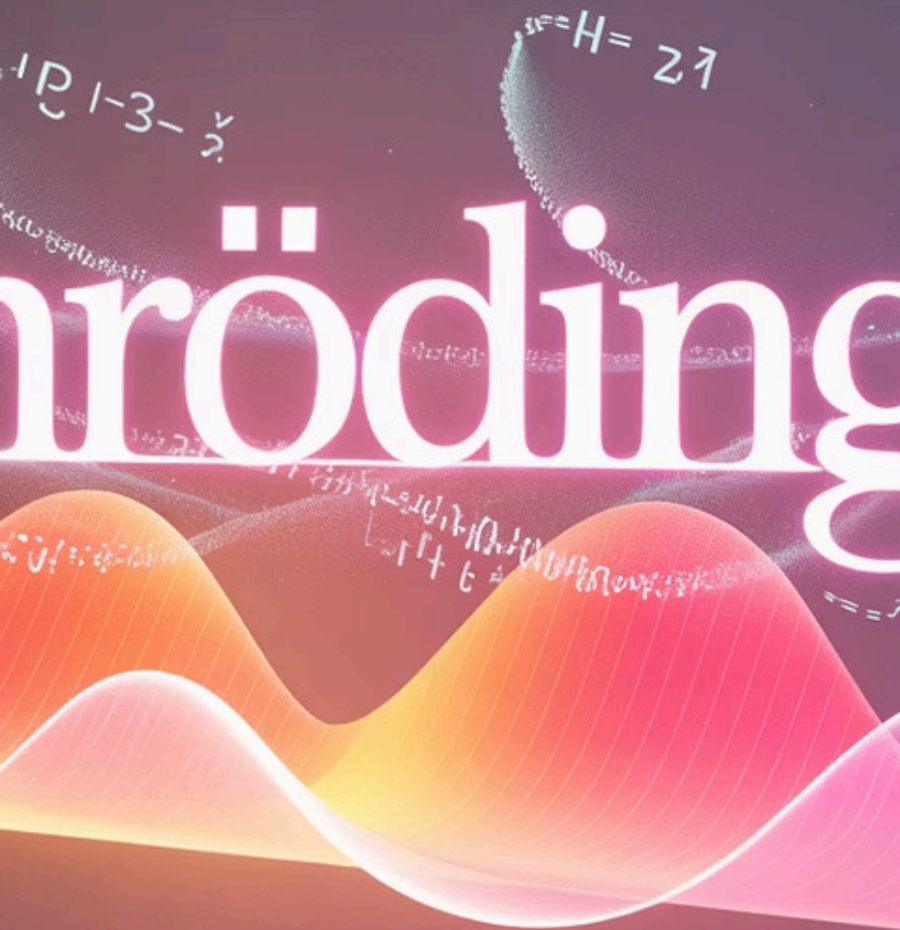


Quantum Tunneling: When Particles Defy Classical Barriers

Welcome to an exploration of one of quantum mechanics' most fascinating phenomena: quantum tunneling. In this presentation, we'll examine how particles can penetrate barriers that would be impossible according to classical physics, and discover the mathematical foundations and real-world applications of this counterintuitive behavior.





The Mathematical Foundation: Schrödinger's Equation

$$i\hbar \frac{\partial \psi(x, t)}{\partial t} = \left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x) \right] \psi(x, t)$$

The time-dependent Schrödinger equation above governs the evolution of quantum wave packets. This fundamental equation describes how a quantum state changes over time, incorporating both kinetic energy (the derivative term) and potential energy $V(x)$.

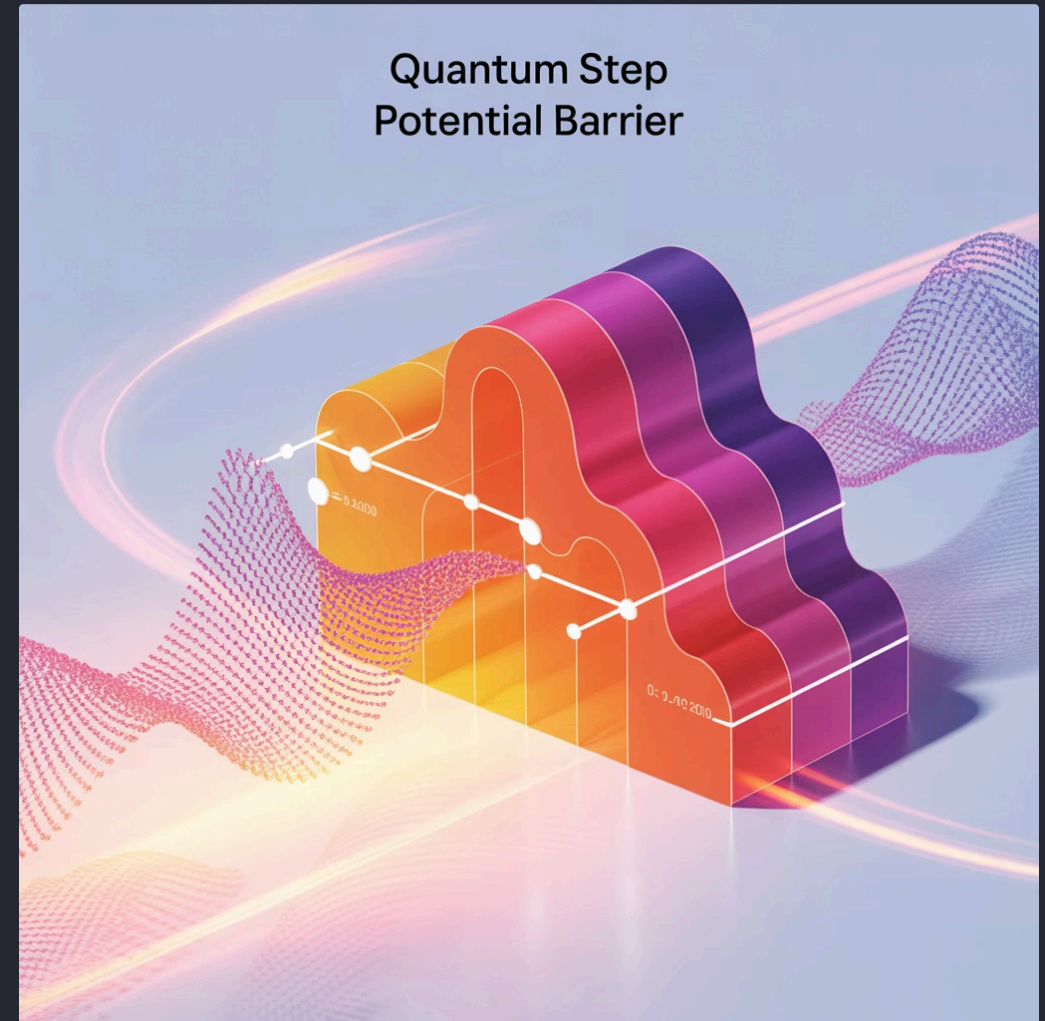
For a wave packet $\psi(x, t)$ initially localized in space, this equation precisely predicts how the probability amplitude will spread and interact with potential barriers in ways that challenge our classical intuition.

The Step Potential Barrier

Our wave packet begins localized on the left side of a step potential barrier, which is mathematically defined as:

$$V(x) = \begin{cases} 0, & x < 0 \\ V_0, & x \geq 0 \end{cases}$$

This represents a sudden jump in potential energy at position $x = 0$. In classical physics, a particle with energy $E < V_0$ would be completely reflected by this barrier, unable to pass through to the other side.



The step potential creates a distinct boundary between two regions with different potential energies, challenging the wave packet to either reflect or transmit through this classically forbidden region.

Quantum Tunneling Phenomenon

Wave-Particle Duality

Unlike classical particles, quantum objects exhibit wave-like properties. The wave packet's probability amplitude $\psi(x, t)$ can extend into and beyond the barrier region.

Probability Density

The squared modulus $|\psi(x, t)|^2$ represents the probability density of finding the particle at position x and time t . Even when $E < V_0$, this value is nonzero beyond the barrier.

Continuity Constraints

The Schrödinger equation requires that $\psi(x, t)$ and its derivative must be continuous at boundaries, which mathematically necessitates some penetration into the barrier.

Although the total energy E of the wave packet is less than the barrier height V_0 , quantum mechanics predicts a nonzero probability amplitude on the other side—a phenomenon that defies classical physics entirely.

Classical vs. Quantum Behavior

Classical Mechanics

- Particles are point-like objects with definite position and momentum
- A particle with $E < V_0$ is completely reflected by the barrier
- The forbidden region is absolutely impenetrable
- Trajectory is deterministic and follows Newton's laws

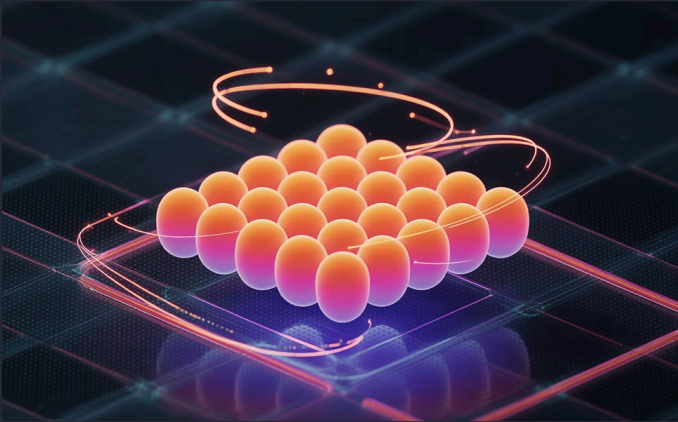
Quantum Mechanics

- Particles exhibit wave-like properties described by wavefunctions
- Part of the wavefunction penetrates and transmits through the barrier
- Probability of tunneling depends on barrier width and height
- Behavior is probabilistic with multiple possible outcomes

This fundamental difference arises from the continuity and differentiability requirements of the Schrödinger equation, which force the wavefunction to extend beyond classical boundaries—creating the tunneling effect that has profound implications for modern technology.

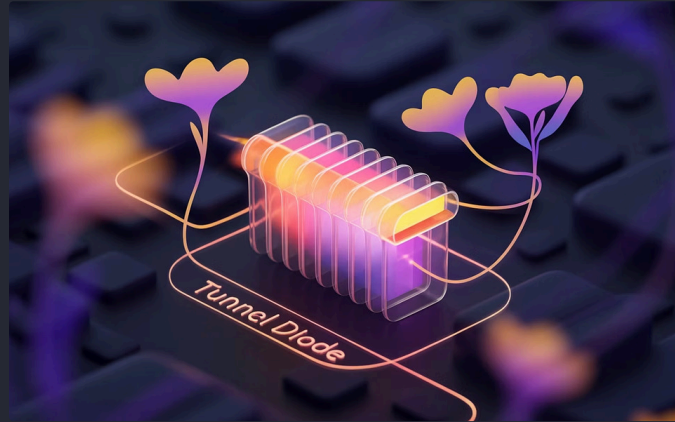


Real-World Applications of Quantum Tunneling



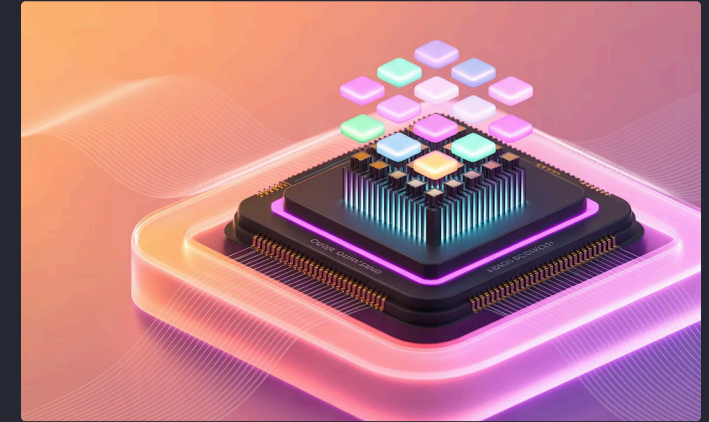
Scanning Tunneling Microscopy

STM uses quantum tunneling of electrons between a sharp metallic tip and a sample surface to image individual atoms with unprecedented resolution, revolutionizing surface science.



Tunnel Diodes

These electronic devices exploit quantum tunneling to achieve extremely fast switching speeds and negative differential resistance, enabling high-frequency applications.



Quantum Computing

Tunneling enables quantum bits to traverse energy barriers, facilitating quantum annealing and other quantum computational processes that classical computers cannot perform.

Quantum tunneling also plays a crucial role in nuclear fusion reactions in stars, alpha decay in radioactive materials, and emerging technologies like tunnel field-effect transistors that promise ultra-low power consumption for future electronics.